

Azure Fundamentals Lab Manual

Hands-on Lab: Virtual Machine, Azure SQL Database, Blob Storage, and a Simple HTML Website

All exercises use the Azure Portal (no command line required).

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Introduction

This lab manual is a guided, hands-on introduction to four core Azure services. Every step is performed in the Azure Portal using point-and-click navigation — no scripting or command-line tools are required. By the end of this lab you will have created a virtual machine, an Azure SQL database, a storage account with image upload, and a simple HTML website hosted on Azure.

What You Will Build

Lab	Outcome
Lab 1: Virtual Machine	A running Windows or Linux VM you can connect to remotely
Lab 2: Azure SQL Database	A cloud SQL database with a server, accessible via Query editor
Lab 3: Blob Storage	A storage account and container with an image file uploaded
Lab 4: Simple HTML Deployment	A static HTML page published to a public URL

Prerequisites

- An active Azure subscription (a free trial or pay-as-you-go subscription both work).
- Permission to create resources in at least one resource group.
- A modern web browser (Edge, Chrome, or Firefox).
- One small image file (JPG or PNG) saved on your computer, for Lab 3.
- A short HTML file, for Lab 4 (a sample is provided in Lab 4).

Warning — Cost Control

Every resource created in this lab manual incurs cost while it is running. Follow the Cleanup Checklist at the end of this document to delete resources when you are finished, especially the virtual machine and the SQL database.

Conventions Used in This Manual

- Numbered steps describe actions to take in the Azure Portal.
- Yellow Note boxes provide helpful context or alternatives.
- Red Warning boxes flag cost or security considerations.
- Text in quotation marks (e.g. “Create a resource”) refers to exact buttons, menu items, or labels in the portal.

Lab 1: Create a Virtual Machine

In this lab, you will deploy a virtual machine (VM) running Windows Server or Ubuntu Linux directly from the Azure Portal.

Step 1: Sign In and Start the Wizard

1. Go to the Azure Portal at portal.azure.com and sign in with your Azure account.
2. In the search bar at the top of the page, type “Virtual machines” and select it from the results.
3. On the Virtual Machines page, select “+ Create”, then choose “Azure virtual machine” from the dropdown.

Step 2: Configure the Basics Tab

4. Under Project details, select your Subscription.
5. For Resource group, select “Create new” and enter a name, for example LabResourceGroup.
6. Under Instance details, enter a Virtual machine name, for example MyLabVM.
7. Choose a Region close to you.
8. For Image, select a Windows Server image (e.g. Windows Server 2022 Datacenter) or Ubuntu Server (e.g. Ubuntu Server 22.04 LTS).
9. Leave the default Size, or select “See all sizes” and choose a small, low-cost size such as B1s for lab purposes.
10. Under Administrator account, enter a Username and a Password (or upload an SSH public key if using Linux). The password must be at least 12 characters and meet complexity requirements.
11. Under Inbound port rules, allow the relevant inbound port: RDP (3389) for Windows, or SSH (22) for Linux.

Step 3: Review and Create

12. Select “Review + create” at the bottom of the page.
13. Wait for validation to pass, then review the configuration summary.
14. Select “Create”.
15. Wait for the deployment to complete (typically 1–3 minutes), then select “Go to resource”.

Step 4: Connect to the VM

16. On the VM Overview page, select “Connect”.
17. For Windows, choose RDP, download the RDP file, and open it with Remote Desktop Connection using the username and password you set.
18. For Linux, choose SSH and follow the connection command shown, using an SSH client such as PuTTY or a terminal.

Note

If RDP or SSH connection fails, confirm the inbound port rule was created and that your network allows outbound connections on that port.

Dashboard

**CreateVm-MicrosoftWindowsServer.WindowsServer-202-20260630102821** | Overview ...
Deployment

x <<     

Overview
Inputs
Outputs
Template

Deployment is in progress
Deployment name : CreateVm-MicrosoftWindowsServer.Windows... Start time : 6/30/2026, 10:33:57 AM
Subscription : Training-2026 Correlation ID : 7d07bc17-39e7-4a06-b741-e8d7173ca192
Resource group : LabResourceGroupVaibhav
Deployment details

Resource	Type	Status	Operation details
 MyLabVMVaibhav	 Microsoft.Compute/virtualMach	Created	Operation details
 mylabvmvaibhav80	 Microsoft.Network/networkInter	OK	Operation details
 MyLabVMVaibhav-nsg	 Network security group	OK	Operation details
 MyLabVMVaibhav-ip	 Public IP address	OK	Operation details
 MyLabVMVaibhav-vnet	 Virtual network	OK	Operation details

Next steps

Dashboard

**CreateVm-MicrosoftWindowsServer.WindowsServer-202-20260630102821** | Overview ...
Deployment

x <<     

Overview
Inputs
Outputs
Template

Your deployment is complete
Deployment name : CreateVm-MicrosoftWindowsServer.Windows... Start time : 6/30/2026, 10:34:05 AM
Subscription : Training-2026 Correlation ID : 7d07bc17-39e7-4a06-b741-e8d7173ca192
Resource group : LabResourceGroupVaibhav
Deployment details

Resource	Type	Status	Operation details
 MyLabVMVaibhav	 Microsoft.Compute/virtualMach	OK	Operation details
 mylabvmvaibhav80	 Microsoft.Network/networkInter	OK	Operation details
 MyLabVMVaibhav-nsg	 Network security group	OK	Operation details
 MyLabVMVaibhav-ip	 Public IP address	OK	Operation details
 MyLabVMVaibhav-vnet	 Virtual network	OK	Operation details

Next steps

- [Set up auto-shutdown](#) Recommended
- [Monitor VM health, performance, and network dependencies](#) Recommended
- [Run a script inside the virtual machine](#) Recommended

 Cos
Get
prev
Set

The screenshot displays a Windows 10 desktop environment. The primary application in the foreground is the **Server Manager** window, which is titled "Server Manager > All Servers". The left-hand navigation pane includes links to "Dashboard", "Local Server", "All Servers" (which is currently selected), and "File and Storage Services". The main content area is divided into two sections: "EVENTS" and "SERVICES".

The **EVENTS** section, titled "All events | 17 total", contains a table with the following data:

Server Name	ID	Severity	Source	Log	Date and Time
MyLabVMVaibhav	24	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:17 AM
MyLabVMVaibhav	23	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:16 AM
MyLabVMVaibhav	24	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:11 AM
MyLabVMVaibhav	24	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:11 AM
MyLabVMVaibhav	23	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:09 AM
MyLabVMVaibhav	23	Warning	Microsoft-Windows-AppModel-State	Application	6/30/2026 5:13:09 AM
MyLabVMVaibhav	1800	Warning	Microsoft-Windows-TPM-WMI	System	6/30/2026 5:10:15 AM

The **SERVICES** section, titled "All services | 234 total", contains a table with the following data:

Server Name	Display Name	Service Name	Status	Start Type
MyLabVMVaibhav	Application Information	AppInfo	Running	Manual (Triggered)
MyLabVMVaibhav	App Readiness	AppReadiness	Running	Manual
MyLabVMVaibhav	AppX Deployment Service (AppXSVC)	AppXSvc	Running	Automatic (Triggered)
MyLabVMVaibhav	Windows Audio	AudioSrv	Running	Automatic

In the background, a Microsoft Edge browser window is open, displaying the Microsoft Azure website. The address bar shows the URL <https://www.bing.com/search?q=azure&cid=065b7f755038...>. The search results include the "Microsoft Azure Official Site" with a link to "Create Your Free Account Today". The page also features a "Developer" section with a "License" section indicating it is "Proprietary for platform, MIT..".

Lab 2: Create an Azure SQL Database

In this lab, you will create a logical SQL server and a single Azure SQL Database, then query it using the built-in Query editor — entirely in the browser.

Step 1: Start the Wizard

19. In the Azure Portal search bar, type “SQL databases” and select it.
20. Select “+ Create”.

Step 2: Configure the Basics Tab

21. Select your Subscription.
22. For Resource group, select the same resource group used in Lab 1, or create a new one.
23. Under Database details, enter a Database name, for example mySampleDatabase.
24. Next to Server, select “Create new”. Enter a unique Server name, choose a Location, select an authentication method (SQL authentication is simplest for a lab), and set a Server admin login and Password. Select “OK”.
25. Leave “Want to use SQL elastic pool?” set to No.
26. Under Compute + storage, select “Configure database”, choose the General Purpose tier with Serverless compute, and select the smallest vCore option to minimize cost. Select “Apply”.
27. Under Backup storage redundancy, choose Locally-redundant backup storage (lowest cost for a lab).

Step 3: Configure Networking

28. Select the “Networking” tab.
29. For Connectivity method, select “Public endpoint”.
30. Set “Allow Azure services and resources to access this server” to Yes.
31. Set “Add current client IP address” to Yes so you can connect from your own machine.

Step 4: Review, Create, and Query

32. Select “Review + create”, wait for validation, then select “Create”.
33. Once deployment finishes, select “Go to resource”.
34. On the database Overview page, select “Query editor (preview)” in the left menu.
35. Sign in using the SQL admin login and password you created.
36. In the query window, run a simple test query, for example: `SELECT @@VERSION;`, then select “Run” to confirm the database is working.

Note

The Query editor runs entirely in the browser, so no database client software is required for this lab.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is logged in as 'vgupta1@presidio.com'. The main content area is titled 'Microsoft.SQLDatabase.newDatabaseNewServer_01da8708f7484dd19b804 | Overview'. The deployment status is 'Your deployment is complete'. The deployment details table lists the following resources:

Resource	Type	Status	Operation details
vaibhavgupta/Vaibhav'sSamp	Microsoft.Sql/servers/databases	Created	Operation details
vaibhavgupta/Default	Microsoft.Sql/servers/connectio	OK	Operation details
vaibhavgupta/Clientip-2026-	Microsoft.Sql/servers/firewallrul	OK	Operation details
vaibhavgupta/AllowAllWindo	Microsoft.Sql/servers/firewallrul	OK	Operation details
vaibhavgupta	Microsoft.Sql/servers	Created	Operation details

The 'Next steps' section includes a 'Go to resource' button. The right sidebar contains links for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

The screenshot displays the Microsoft Azure portal interface for the 'Vaibhav'sSampleDatabase (vaibhavgupta/Vaibhav'sSampleDatabase) | Query editor (preview)'. The left sidebar shows the 'Query editor (preview)' tab selected. The main content area shows the 'SQL query 1' editor with the following query:

```
SELECT @@VERSION
```

The 'Results' tab is active, showing a single row of results:

ABR
Microsoft SQL Azure (RTM) - 12.0.2000.8 Apr 14 2026 20:27:12 Copyright (C) 2025 Microsoft Corporation

The status bar at the bottom indicates 'Succeeded (0 sec 221 ms)' and 'Columns: 1 Rows: 1'.

Lab 3: Blob Storage and Image Upload

In this lab, you will create a storage account, add a container, and upload an image file as a blob.

Step 1: Create a Storage Account

37. In the Azure Portal search bar, type “Storage accounts” and select it.
38. Select “+ Create”.
39. Select your Subscription and the resource group from Lab 1.
40. Enter a globally unique Storage account name (lowercase letters and numbers only, e.g. mylabstorage2026).
41. Choose a Region close to you.
42. Leave Performance set to Standard and Redundancy set to Locally-redundant storage (LRS) for a low-cost lab setup.
43. Select “Review”, then “Create”, and wait for deployment to finish.
44. Select “Go to resource”.

Step 2: Create a Container

45. In the left menu of the storage account, under Data storage, select “Containers”.
46. Select “+ Container”.
47. Enter a name, for example images (lowercase letters, numbers, and dashes only).
48. Leave the default Public access level (Private), then select “Create”.

Step 3: Upload an Image

49. Select the new container to open it.
50. Select “Upload”.
51. Browse to the image file on your computer (JPG or PNG) and select it.
52. Select “Upload” to complete the operation.
53. Select the uploaded blob in the list to view its properties and copy its URL.

Note

If the container's public access level is Private, the blob URL will not be publicly viewable in a browser. To make an individual image viewable via its URL, set the container access level to “Blob” (anonymous read access for blobs only) when creating it, or change it afterward under container Settings > Access level.

vaibhavlabstorage2026_1782799572233 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : vaibhavlabstorage2026_1782799572233 Start time : 6/30/2026, 11:37:22 AM

Subscription : Training-2026 Correlation ID : 26f2ccea-13d0-4efd-89e1-c45629c962fe

Resource group : LabResourceGroupVaibhav

Deployment details

Resource	Type	Status	Operation details
vaibhavlabstorage2026/defa	Microsoft.Storage/storageAccou	OK	Operation details
vaibhavlabstorage2026/defa	Microsoft.Storage/storageAccou	OK	Operation details
vaibhavlabstorage2026	Storage account	OK	Operation details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

vgupta1@presidio.com

Presidio Networked Solution...

Dashboard > vaibhavlabstorage2026_1782799572233

vaibhavimages

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Auth

Show

images.jpeg

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL

https://vaibhavlabstorag...

LAST MODIFIED

6/30/2026, 11:42:29 AM

CREATION TIME

6/30/2026, 11:42:29 AM

VERSION ID

-

TYPE

Block blob

SIZE

15.62 KiB

ACCESS TIER

Hot (Inferred)

ACCESS TIER LAST MODIFIED

N/A

ARCHIVE STATUS

-

REHYDRATE PRIORITY

-

SERVER ENCRYPTED

true

ETAG

0x8DED66E90885021

VERSION-LEVEL IMMUTABILITY POLICY

Disabled

CACHE-CONTROL

CONTENT-TYPE

image/jpeg

CONTENT-MD5

YckAHZ3qtUF+jlZU1G4NzQ==

CONTENT-ENCODING

CONTENT-LANGUAGE

CONTENT-DISPOSITION

LEASE STATUS

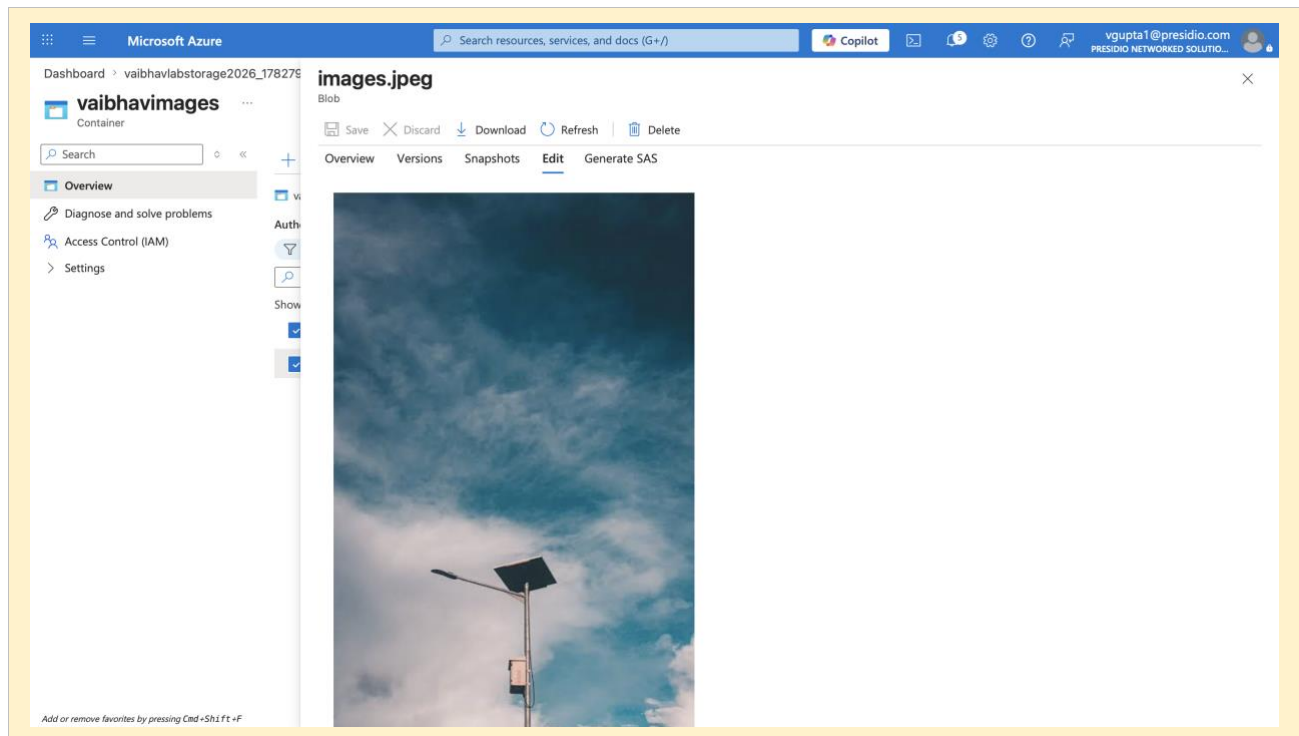
Unlocked

LEASE STATE

Available

LEASE DURATION

-



Lab 4: Deploy a Simple HTML Website

In this lab, you will publish a basic HTML page to the public internet using the static website feature of the storage account you created in Lab 3 — all from the portal, with no command-line deployment tools.

Step 1: Prepare a Sample HTML File

Create a text file named `index.html` on your computer with the following content, then save it:

```
<!DOCTYPE html> <html> <head><title>My Azure Lab Site</title></head> <body>   <h1>Hello from  
Azure Blob Storage!</h1>   <p>This page was deployed using static website hosting.</p> </body>  
</html>
```

Step 2: Enable Static Website Hosting

54. Open the storage account created in Lab 3.
55. In the left menu, under Data management, select “Static website”.
56. Select “Enabled”.
57. For Index document name, enter `index.html`.
58. Select “Save”.
59. Azure displays a Primary endpoint URL once saved — copy this URL; it is the public address of your site.

Step 3: Upload the HTML File

60. After saving, return to the Static website page and select the link next to “\$web container” (Azure automatically creates a container named \$web for static content).
61. Select “Upload”.
62. Browse to your `index.html` file and select it.
63. Select “Upload” to complete the operation.

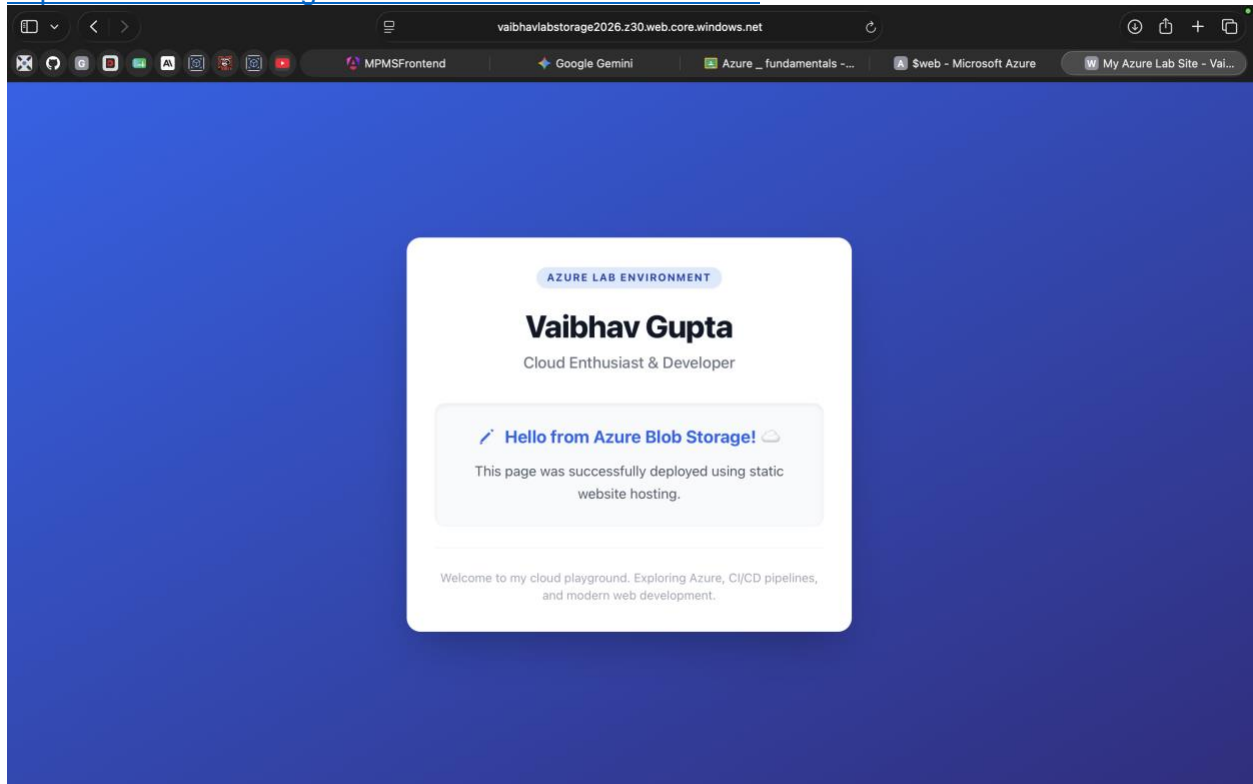
Step 4: View the Site

64. Return to the Static website page in the left menu of the storage account.
65. Select or copy the Primary endpoint URL and open it in a new browser tab.
66. Confirm that your HTML page renders correctly.

Note

The \$web container is automatically configured for public, anonymous read access when static website hosting is enabled, so your page is reachable by anyone with the URL.

<https://vaibhavlabstorage2026.z30.web.core.windows.net/>



Cleanup Checklist

Azure resources continue to incur charges until they are deleted. When you have finished the lab, clean up resources to avoid unnecessary cost.

Option A: Delete the Entire Resource Group (Recommended)

67. In the Azure Portal search bar, type “Resource groups” and select it.
68. Select the resource group used throughout this lab (e.g. LabResourceGroup).
69. Select “Delete resource group”.
70. Type the resource group name to confirm, then select “Delete”.

Deleting the resource group removes the VM, SQL server and database, storage account, and all associated resources (disks, network interfaces, public IP addresses) in one step.

Dashboard

LabResourceGroupVaibhav

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Resources

Recommendations

Filter for any field...

Type equals all

Location equals all

Add filter

Name	Type	Location
MyLabVMVaibhav	Virtual machine	South India
MyLabVMVaibhav-ip	Public IP address	South India
MyLabVMVaibhav-nsg	Network security group	South India
MyLabVMVaibhav-vnet	Virtual network	South India
mylabvmvaibhav80	Network interface	South India
MyLabVMVaibhav_OsDisk_1_cf5eb3a1e63748a59c9d1a25b41b7f0	Disk	South India
Vaibhav'sSampleDatabase (vaibhavgupta/Vaibhav'sSampleDatabase)	SQL database	South India
vaibhavgupta	SQL server	South India
vaibhavlstorage2026	Storage account	South India

Showing 1 - 9 of 9. Display count: 100

Add or remove favorites by pressing **Cmd+Shift+F**

Dashboard

LabResourceGroupVaibhav

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Resources

Recommendations

Filter for any field...

Type equals all

Showing 1 - 9 of 9. Display count: 100

Add or remove favorites by pressing **Cmd+Shift+F**

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

LabResourceGroupVaibhav

10 resources being deleted across 1 subscriptions, 2 resource groups

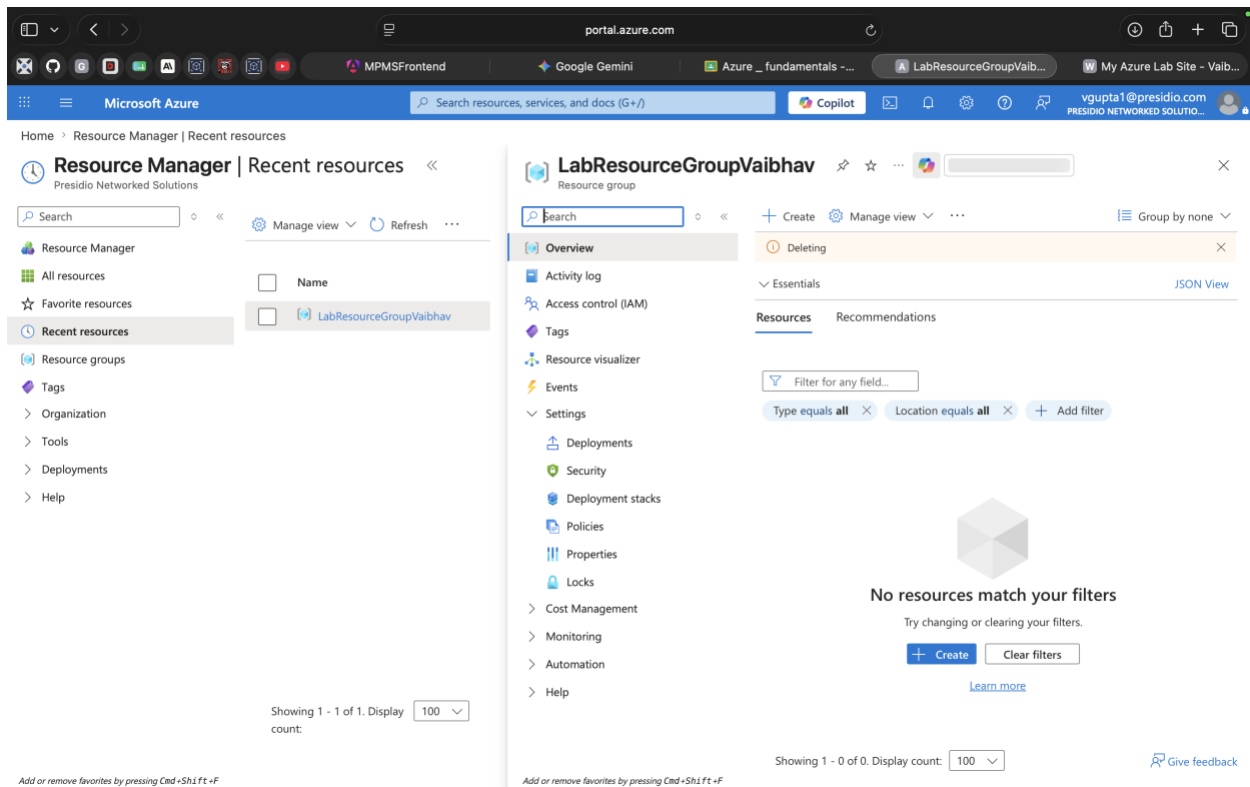
All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
master (vaibhavgupta/master)	SQL database	Training-2026	LabResourceGroupV...
MyLabVMVaibhav	Virtual machine	Training-2026	LabResourceGroupV...
MyLabVMVaibhav-ip	Public IP address	Training-2026	LabResourceGroupV...
MyLabVMVaibhav-nsg	Network security group	Training-2026	LabResourceGroupV...
MyLabVMVaibhav-vnet	Virtual network	Training-2026	LabResourceGroupV...
mylabvmvaibhav80	Network interface	Training-2026	LabResourceGroupV...
MyLabVMVaibhav_OsDisk_1_cf5eb3a1e63748a59c	Disk	Training-2026	LABRESOURCEGROU...
Vaibhav'sSampleDatabase (vaibhavgupta/Vaibhav'sSampleDatabase)	SQL database	Training-2026	LabResourceGroupV...
vaibhavgupta	SQL server	Training-2026	LabResourceGroupV...
vaibhavlstorage2026	Storage account	Training-2026	LabResourceGroupV...

Enter resource group name to confirm deletion *

LabResourceGroupVaibhav

Delete Cancel



Option B: Delete Resources Individually

- Virtual machine: open the VM, select “Delete”, and also delete the associated disk, network interface, public IP, and network security group shown on the VM's Overview page.
- SQL Database and server: open the SQL server resource, select “Delete” (this also offers to delete databases on that server).
- Storage account: open the storage account, select “Delete”, and confirm by typing the account name.

Warning — Cost Control

A virtual machine continues to bill for compute time while running, and for attached disks even when stopped. Always delete or fully deallocate resources you no longer need.

Summary

In this lab manual you created and configured four foundational Azure resources entirely through the Azure Portal: a virtual machine, an Azure SQL Database, a Blob Storage container with an uploaded image, and a static HTML website. These exercises cover compute, database, and storage — three of the core pillars of the Azure platform — and provide a foundation for more advanced labs involving networking, security, and automation.

Further Reading

[Create a Windows VM in the Azure portal](#)

[Create a single Azure SQL database \(portal\)](#)

[Upload, download, and list blobs \(Azure portal\)](#)

[Host a static website in Azure Storage](#)